

CONFIDENTIAL - CP Group Internal Use Only

iCENTRAL WAR ROOM & TRUE VISION WAR ROOM (PMS)

Infrastructure, Security & Technical Documentation

With Data Security Policy Compliance

Field	Value
Version	3.0
Date	January 2026
Status	Production
Classification	Internal - Confidential

Business Strategy Development Office (BSDO) CP Group - Chairman Office

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1. Executive Summary

The iCentral War Room ecosystem consists of two integrated platforms designed to provide strategic intelligence and operational visibility across CP Group's business units. This documentation focuses primarily on the True Vision War Room system while providing context for how it integrates with the broader iCentral platform.

Key Integration Points

SSO Integration: True Vision War Room serves as the authentication provider for the entire ecosystem

Security Compliance: Both systems comply with CP Group Data Security Procedure (Effective April 1, 2024)

Primary Focus: True Vision PMS, BSC, and Analytics modules with enterprise security features

2. System Architecture & SSO Flow

The following diagram illustrates how the two systems work together through Single Sign-On (SSO) authentication.

2.1 Authentication Flow Diagram



3. iCentral War Room Overview

The iCentral War Room serves as the central intelligence hub for CP Group, providing macro-level business insights. It authenticates users via SSO from True Vision War Room.

3.1 Core Capabilities

Feature	Description
Macro View	Global economy indicators (GDP, Inflation, Exchange Rates)
Capital Market	Real-time stock prices via Yahoo Finance API (CPF, TRUE, CPALL)
Overseas Market	NASDAQ, Hang Seng, ITOCHU, PingAn tracking
Distribution Maps	Interactive maps (World, Thailand, China) via amCharts
BU Access	Gateway to 20+ business unit dashboards

3.2 Technical Stack

Component	Technology
Frontend	React 18 + Material-UI + amCharts + D3.js + Three.js
Backend	Node.js + Express.js
Database	MySQL 8.0 (Database: CIW)
Authentication	SSO via True Vision War Room (JWT)

4. True Vision War Room (Primary Focus)

True Vision War Room is the SSO authentication provider and primary operational platform, offering Project Management (PMS), Balanced Scorecard (BSC), and AI-powered Analytics modules.

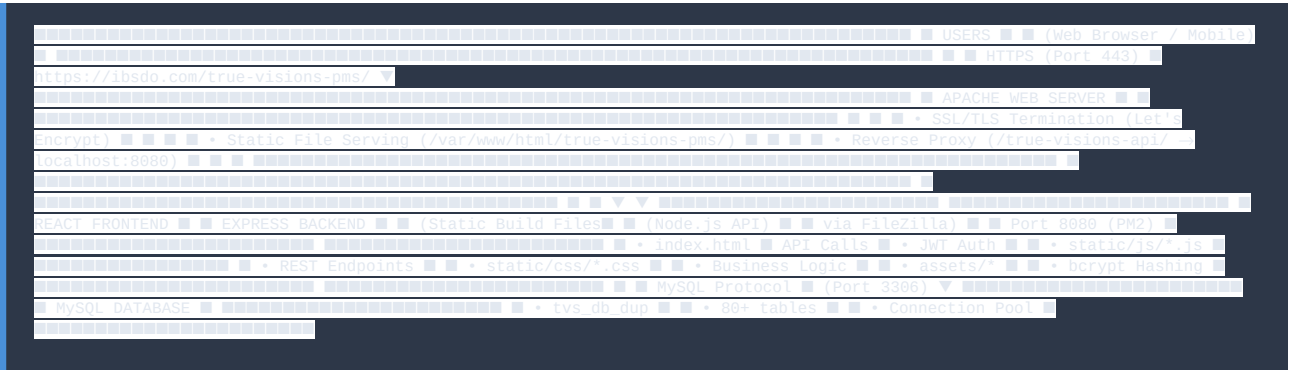
4.1 System Modules

PMS - Project Management	BSC - Balanced Scorecard	Analytics - AI Forecasting
Project Tracking	Strategy Map	95% AI Accuracy
Team & ROI Analysis	4 Perspectives	Revenue Trends
Action Plans	KPI Summary	Churn Analysis
Timeline Management	Performance Metrics	Sales Heatmaps

4.2 Technical Stack

Component	Technology
Frontend	React 18 + Material-UI + ApexCharts + Redux
Frontend Deployment	Static build files deployed via FileZilla (SFTP) to Apache web server
Frontend URL	https://ibsdo.com/true-visions-pms/
Backend	Node.js + Express.js (Internal Port 8080, managed by PM2)
Backend URL	https://ibsdo.com/true-visions-api/ (via Apache Reverse Proxy)
Web Server	Apache 2.4.41 (SSL termination + reverse proxy)
SSL	Let's Encrypt (HTTPS, auto-renewal)
Database	MySQL 8.0 (Database: tvs_db_dup)
Authentication	JWT (30 min) + bcrypt (12 rounds) + SSO Provider
BI Integration	Power BI Embedded (BSC + Analytics)
Email Service	Nodemailer + Brevo SMTP (OTP delivery)
Process Manager	PM2 (auto-restart on crash)

4.2.1 Deployment Architecture

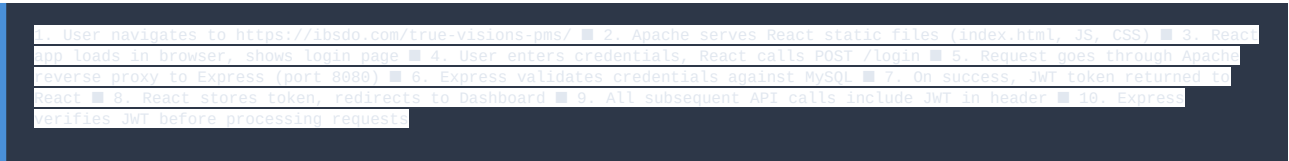


4.2.2 Frontend Deployment Process

The React frontend is built locally and deployed as static files via FileZilla:



4.2.3 Request Flow



4.3 Enterprise Security Features (SSO Provider)

True Vision War Room implements comprehensive security controls that are shared across all connected systems via SSO:

Security Feature	Implementation
Password Hashing	bcrypt with 12 salt rounds
JWT Token Expiry	30 minutes
Account Lockout	3 failed attempts
Lockout Duration	10 minutes (auto-unlock)
Password History	Last 5 passwords prevented
Password Expiry	12 months mandatory change
Single Session	Yes - per user enforcement
Audit Logging	Complete activity trail
OTP Recovery	6-digit email OTP (10 min validity)
Session Timeout	30 minutes inactivity

5. Data Security Policy Reference

Reference Document: Data Security Procedure of Central War Room **Effective Date:** April 1, 2024 | Revision: 1 **Approved By:** Mr. Soopakij Chearavanont, Chairman of CP Group

5.1 Policy Objectives

The Central War Room is established according to the direction of the Chairman for management and monitoring of CP Group companies. Key user access levels:

- **Level 1.1.1:** Senior Chairman, Chairman, CEO of CP Group - Full access to all information
- **Level 1.1.2:** C-Level executives - Access to own business group (requires Chairman approval)
- **Level 1.1.3:** Assigned users - Specified access (requires multi-level approval chain)
- **Level 1.1.4:** System Administrators and Working Teams - Chairman approval required

5.2 Key Policy Requirements

5.2.1 Access Control (Policy Section 5.1)

- All users must sign NDA for confidentiality
- Access Authorization Matrix defines user permissions
- Data copying/distribution requires Chairman approval

5.2.2 Password Management (Policy Section 5.2.3.6)

- Password not displayed during entry
- Password change required every 12 months
- Cannot reuse last 5 passwords
- Minimum 8 characters with letters, numbers, special characters
- Account lockout after 5 failed attempts
- MAC Address control for device authorization

5.2.3 Session Management (Policy Section 5.2.3.9)

- Auto-lock after 30 minutes of inactivity
- Password required to resume session

5.2.4 User Access Review (Policy Section 5.2.4)

- User list review every 6 months
- Inactive users (6+ months) must be reviewed
- Immediate deactivation upon resignation/transfer

6. Security Implementation Compliance Checklist

This checklist maps the implemented security features against the Data Security Policy requirements.

Legend: ✓ = Implemented, ■ = Partial, X = Not Implemented

PASSWORD SECURITY

Status	Policy Requirement	Implementation	Policy Ref
✓	Password not displayed during entry	Input type=password	5.2.3.6
✓	Password change every 12 months	12-month expiry + auto-force	5.2.3.6
✓	Cannot reuse last 5 passwords	Password history table	5.2.3.6
✓	Min 8 chars with mixed characters	Regex validation	5.2.3.6
✓	Account lockout after failed attempts	3 attempts (policy: 5)	5.2.3.6
✓	Auto-unlock after lockout period	10 minutes	5.3.1.9
✓	Force password change on first login	Implemented	5.2.3.4
■	MAC Address device control	Not yet implemented	5.2.3.6

SESSION MANAGEMENT

Status	Policy Requirement	Implementation	Policy Ref
✓	Session timeout (30 min inactivity)	JWT 30-min expiry	5.2.3.9
✓	Single session per user	Token invalidation on new login	5.2.3.2
✓	Password required to resume	Re-authentication required	5.2.3.9

USER MANAGEMENT

Status	Policy Requirement	Implementation	Policy Ref
✓	Individual user IDs required	Unique usernames enforced	5.2.3.1
✓	No shared/group usernames	BINARY comparison	5.2.3.2
✓	User expiration dates	Start/end date fields	5.2.3.6
■	Access review every 6 months	Manual process required	5.2.4.1

AUDIT & LOGGING

Status	Policy Requirement	Implementation	Policy Ref
✓	Login attempt logging	Full audit trail	5.2.3.6
✓	Record all system access attempts	IP, User Agent, Timestamp	5.2.3.6
✓	Page access logging	User ID, Path, Title tracked	5.2.3.6
✓	Password change logging	Hash stored in history	5.2.3.6

SYSTEM SECURITY

Status	Policy Requirement	Implementation	Policy Ref
✓	Warning message on login page	CP Group ownership displayed	5.2.3.5
✓	Secure password storage	bcrypt 12 rounds	5.4.2.1
✓	Encrypted password transmission	HTTPS/SSL	5.2.3.8
✓	Password reset verification	OTP via email (6-digit)	5.2.3.4
✓	Default password change required	Force change on first login	5.4.2.2

6.1 Compliance Summary

Category	Count
Fully Implemented (✓)	19
Partially Implemented (■)	2
Not Implemented (✗)	0
Compliance Rate	90%+

6.2 Action Items for Full Compliance

1. **MAC Address Control:** Implement device registration and validation system (Policy 5.2.3.6)
2. **6-Month Access Review:** Establish automated user access review process (Policy 5.2.4.1)

7. Infrastructure Details

7.1 Hosting Environment

Hosted on True IDC - Thailand's Premier Data Center

Parameter	Value
VDC Name	IDC2023_5891_True Visions Group_VDC_New
Organization	True_Visions_Group
Host	vm-az01.trueidc.com
CPU	16 GHz Allocated
Memory	16 GB Used / 20 GB Allocated
Storage	66 GB Used / 200 GB Allocated

7.2 Network Configuration

Service	Port	Description
HTTPS	443	SSL/TLS encrypted traffic (all public access)
HTTP	80	Redirects to HTTPS
True Vision Backend	8080 (internal)	Node.js API (proxied via Apache at /true-visions-api/)
MySQL	3306 (internal)	Database (internal only, not exposed)

7.2.1 Public URLs

System	URL
True Vision PMS Frontend	https://ibsd0.com/true-visions-pms/
True Vision PMS API	https://ibsd0.com/true-visions-api/

Note: All public traffic goes through HTTPS (Port 443) via Apache. Internal service ports (8080, 3306) are NOT exposed to the internet. Apache handles SSL termination and reverse proxying.

7.3 Server Directory Structure

```
server: ibsd0.com (Ubuntu Server)
├── /var/www/html/
│   ├── true-visions-pms/ ← React Build Files (deployed via FileZilla)
│   ├── index.html
│   ├── favicon.ico
│   ├── manifest.json
│   ├── asset-manifest.json
│   ├── robots.txt
│   ├── assets/
│   ├── images, icons
│   ├── true-logo.png
│   ├── dashboard.png
│   ├── ...
│   ├── static/
│   ├── js/ ← Compiled JavaScript
│   ├── main.xxxxx.js
│   ├── css/ ← Compiled CSS
│   ├── main.xxxxx.css
│   └── /home/kittinv/
│       ├── true-vision-warroom/
│       ├── server/ ← Node.js Backend (Managed by PM2)
│       ├── app.js ← Main application
│       ├── package.json ← Dependencies
│       ├── package-lock.json
│       ├── .env ← Configuration
│       └── node_modules/ ← Installed packages
```

7.4 Apache Configuration

```
<VirtualHost *:443>
    ServerName ibsdo.com
    DocumentRoot /var/www/html
    # SSL Configuration (Let's Encrypt)
    SSLCertificateFile /etc/letsencrypt/live/ibsdo.com/fullchain.pem
    SSLCertificateKeyFile /etc/letsencrypt/live/ibsdo.com/privkey.pem
    # Reverse Proxy for True Vision API
    ProxyPass "/true-visions-api/" "http://127.0.0.1:8080/"
    ProxyPassReverse "/true-visions-api/" "http://127.0.0.1:8080/"
    <Location "/true-visions-api/">
        Require all granted
    </Location>
</VirtualHost>
```

7.5 Database Architecture

Parameter	True Vision
Database Name	tvb_db_dup
Server	localhost:3306
Connection Type	Pool (10 max)
Keep-Alive	Yes
Auto-reconnect	Yes

7.6 PM2 Process Management

```
# View running processes
pm2 list
# Start application
pm2 start app.js --name "true-visions-pms-api"
# View logs
pm2 logs true-visions-pms-api
# Restart application
pm2 restart true-visions-pms-api
# Stop application
pm2 stop true-visions-pms-api
# Save process list (for auto-start on reboot)
pm2 save
# Setup auto-start on server reboot
pm2 startup
```

7.7 Environment Configuration (.env)

```
Database DB_HOST=localhost DB_USER=ibsdo DB_PASSWORD=***** DB_NAME=tvb_db_dup DB_PORT=3306
# Server PORT=80
NODE_ENV=production
# Security
JWT_SECRET=your-secure-secret-key
ADMIN_RESET_KEY=admin-reset-key
# Email (OTP)
SMTP_HOST=smtp-relay.brevo.com SMTP_PORT=587 SMTP_USER=your-smtp-user SMTP_PASS=your-smtp-password SMTP_FROM="True Vision War Room" <noreply@company.com>
```

Appendix A: Quick Reference Card

Ports

Service	Port	Access
HTTPS	443	Public
HTTP (redirect)	80	Public (redirects to 443)
Backend API	8080	Internal only (via Apache reverse proxy)
MySQL	3306	Internal only

URLs

Environment	URL
Production Frontend	https://ibsd0.com/true-visions-pms/
Production API	https://ibsd0.com/true-visions-api/

Security Settings

Setting	Value
Session Timeout	30 minutes
Failed Login Limit	3 attempts
Lockout Duration	10 minutes
Password Min Length	8 characters
Password History	5 passwords
OTP Validity	10 minutes
JWT Expiry	30 minutes

Key Commands

```
# Check backend status pm2 status # View logs pm2 logs true-visions-pms-api --lines 50 # Restart backend pm2 restart true-visions-pms-api # Reload Apache sudo systemctl reload apache2 # Test MySQL connection mysql -u ibsd0 -p tvs db.dur
```

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